Unit 3 Part 1 Exam

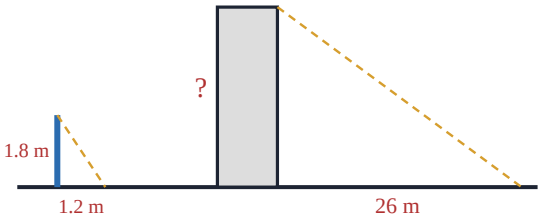
Math 9E Part 1 file · Lesson 3.1 + unit conversions 8 questions · 23 marks Name: _____

The most important questions predicted from this worksheet file. Show all your work. Answers are on the last page.

Question 1

[3 marks]

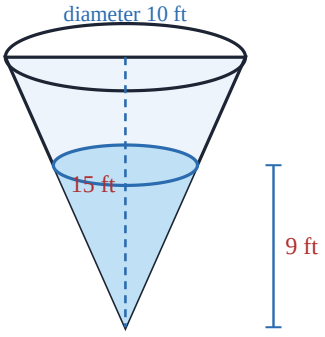
A 1.8 m pole casts a 1.2 m shadow. At the same time a building casts a 26 m shadow. (a) Find the building's height. (b) Later, the building's shadow is 39 m. How long is the pole's shadow then?



Question 2

[3 marks]

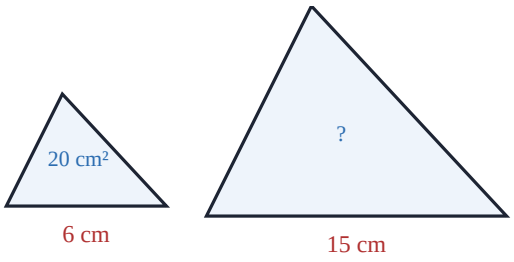
A conical tank (point down) is 15 ft deep with top diameter 10 ft. It is filled to a depth of 9 ft. Find (a) the area of the water surface and (b) the volume of water.



Question 3

[3 marks]

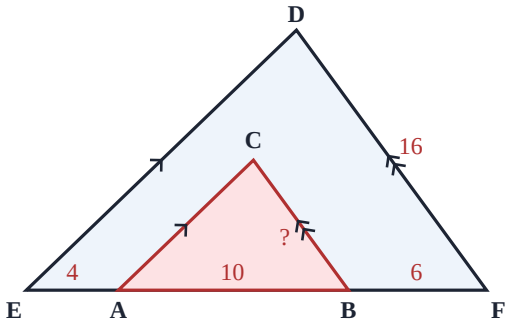
Corresponding sides of two similar triangles are 6 cm and 15 cm. The small triangle has area 20 cm². Find the area of the large one.



Question 4

[4 marks]

In ΔEFD, points A and B lie on EF with EA = 4, AB = 10 and BF = 6. Point C is inside the triangle with AC ∥ ED and BC ∥ FD. FD = 16. (a) Prove ΔABC ∼ ΔEFD. (b) Find BC.



Question 5

[2 marks]

Convert 7.2×10^{-4} km to μm, and 0.003 Gm to mm. Write both answers in scientific notation.

Question 6

[3 marks]

A farm has an area of 2.5 km². Convert this to (a) m², (b) hectares (1 ha = 10 000 m²), (c) acres.

Question 7

[3 marks]

1 ft = 30.48 cm. (a) How many cm^2 are in 1 ft^2? (b) A room is 12 ft × 15 ft. Find its area in m^2, to the nearest hundredth.

Question 8

[2 marks]

A doctor orders 0.6 g of syrup medicine. The label reads 250 mg in 5 mL. How many mL should be given? How many teaspoons is that (1 tsp = 5 mL)?

ANSWER KEY

Q1 (a) $H/26 = 1.8/1.2 \Rightarrow$ **$H = 39$ m**. (b) Now height = shadow (39 m : 39 m), so the pole's shadow = **1.8 m**.

Q2 $r = 5 \times 9/15 = 3$ ft. (a) $\pi(3^2) =$ **$9\pi \approx 28.27$ ft²**. (b) $\frac{1}{3}\pi(9)(9) =$ **$27\pi \approx 84.82$ ft³**.

Q3 $k = 15/6 = 2.5 \Rightarrow k^2 = 6.25$. Area $= 20 \times 6.25 =$ **125 cm²**.

Q4 (a) $AC \parallel ED \Rightarrow \angle CAB = \angle DEF$ (corresponding angles). $BC \parallel FD \Rightarrow \angle CBA = \angle DFE$. So $\Delta ABC \sim \Delta EFD$ (AA). (b) $EF = 4 + 10 + 6 = 20$, $k = AB/EF = 10/20 = 1/2 \Rightarrow BC = \frac{1}{2} \cdot FD =$ **8**.

Q5 7.2×10^{-4} km = 0.72 m = **7.2×10^5 µm**. 0.003 Gm = 3×10^6 m = **3×10^9 mm**.

Q6 1 km² = $(1000\text{ m})^2 = 10^6$ m². (a) **2 500 000 m²** (b) ÷ 10 000 = **250 ha** (c) × 2.47 = **617.5 acres**.

Q7 (a) $30.48^2 \approx$ **929.03 cm²** (b) 180 ft² × 0.3048^2 m² ≈ **16.72 m²**.

Q8 0.6 g = 600 mg. $600 \times 5/250 =$ **12 mL = 2.4 tsp**.

WHERE EACH QUESTION COMES FROM

Q1	Example 3 and Practice 3.1 #4 (flagpole, Delbert)
Q2	Practice 3.1 #5 (conical tank)
Q3	Practice 3.1 #7 ★★★+ (area ratio)
Q4	Practice 3.1 #9 (prove similar, find x)
Q5	Practice 3.2 #1–2 (metric prefixes, scientific notation)
Q6	Practice 3.2 #2f (km² to acres)
Q7	Practice 3.2 #2g (ft² to metric area)
Q8	Practice 3.2 #3–4 (medicine dose)